

HEALTHCONNECT EXPERIENCE LAB

WEEK 7 CROSS-TRACK VALIDATION & ACCEPTANCE MATRIX

Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Document Control

Field	Detail
Project	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Programme	AnalystLab Africa Experience Lab
Workstream	Project Management
Phase	Week 7 — Testing, Refinement & End-to-End Validation
Document Version	1.1
Document Status	Controlled PM Validation Artifact
Owner	Project Management
Validation Basis	Week 7 workstream outputs, documented handoffs, interface dependencies, test evidence, receiving-track validation and acceptance status

1. PURPOSE

This matrix provides the controlled record of cross-track validation across the HealthConnect workstreams.

Its purpose is to establish whether a material output produced by one track has successfully crossed its intended interface and is:

- Available to the receiving track;
- Consistent with the required definition, structure or contract;
- Usable for its intended downstream purpose;
- Objectively validated where validation is required;
- Subject to corrective action and re-test where a gap is identified; and
- Formally accepted, conditionally accepted, pending or otherwise dispositioned.

The governing control: an output is not considered successfully integrated merely because it has been produced or communicated. Cross-track acceptance requires evidence that the receiving context can use the output for its intended downstream purpose.

This matrix therefore sits between workstream output completion and integrated solution acceptance.

2. CONTROL SCOPE

Cross-track validation is applied only where a material dependency exists between workstreams. For HealthConnect, the principal validation interfaces are:

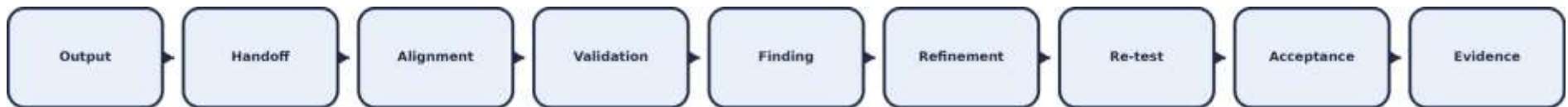
1. Data Analytics ↔ Data Science
2. Data Science ↔ ML Engineering
3. ML Engineering ↔ Operational Decision Pathway
4. Decision Pathway ↔ Intervention Workflow
5. Generative AI ↔ HealthConnect User/Support Workflow
6. All Workstreams ↔ Integrated HealthConnect Solution

A direct ML Engineering ↔ Generative AI technical interface is not treated as established, because no material technical handoff between those tracks has been evidenced.

Where no actual interface exists, the matrix records the position rather than manufacturing an integration requirement.

3. VALIDATION CONTROL MODEL

Each material interface is assessed through the following control sequence:



Step	Question
1. Output	What was produced by the source track?
2. Handoff	Was the output transferred or made available to the receiving context?
3. Alignment	Are the relevant definitions, structures, inputs or assumptions compatible?
4. Validation	Was the receiving-side use or interface objectively tested?
5. Finding	What did the validation establish?
6. Refinement	Was corrective action required?

Step	Question
7. Re-test	Was the affected condition subsequently verified?
8. Acceptance	Has the receiving context accepted the output for its intended use?
9. Evidence	What controlled evidence supports the current status?

This sequence prevents the project from treating communication as integration or component testing as cross-track acceptance.

4. CROSS-TRACK STATUS DEFINITIONS

Status	Control Meaning
Accepted	Required validation is complete and the receiving context has accepted the output for its intended downstream use.
Conditionally Accepted	The output is usable within a defined scope, subject to documented limitations or remaining conditions.
Validated — Acceptance Pending	Objective validation has been performed, but receiving-side acceptance has not yet been formally established.
Pending Validation	The output/interface exists or is expected, but required cross-track validation has not yet been completed.
Blocked	Validation cannot proceed because a material dependency or external constraint prevents execution.
Not Evidenced	Available project evidence is insufficient to establish the claimed handoff or validation state.
Not Applicable	No material interface exists within the assessed project scope.

5. CROSS-TRACK VALIDATION MATRIX

Each row records the controlled acceptance state of one material cross-track interface, condensed to its essential evidence.

ID	Interface	Actual Evidence Position	Acceptance Status
CTV-001	Analytics → DS	Dashboard/EDA evidence exists; formal downstream incorporation or disposition by DS is not fully evidenced	Validated — Acceptance Pending
CTV-002	Analytics → PM	Dashboard available; formal KPI reconciliation remains outstanding	Pending Validation
CTV-003	Analytics → PM	Material filter/interaction behaviour remains partially unverified	Pending Validation
CTV-004	DS → ML	ML interface capability demonstrated; actual final DS-selected candidate has not been evidenced as integrated	Pending Validation

ID	Interface	Actual Evidence Position	Acceptance Status
CTV-005	DS → ML	33-feature compatibility remains unverified	Pending Validation
CTV-006	DS → PM	5-fold ROC-AUC 0.6771±0.0090; ROC-AUC 0.6668; recall 0.6742; train/test gap 0.0571; acceptance implications remain unresolved	Validated — Acceptance Pending
CTV-007	DS → PM	≈0.83 recall / ≈0.56 precision observed in the independent threshold test; final threshold not accepted	Validated — Acceptance Pending
CTV-008	ML → Operational Decision	Scoring validated; no accepted score-to-decision pathway yet demonstrated	Pending Validation
CTV-009	Decision → Intervention	End-to-end mapping remains undefined/unvalidated	Pending Validation
CTV-010	ML → PM/DS	Explainability capability demonstrated within the assessed ML scope	Conditionally Accepted
CTV-011	ML → PM/DS	Candidate recall 0.638 vs baseline recall 0.615	Validated — Acceptance Pending
CTV-012	GenAI → HealthConnect	Grounding tested within selected scenarios; broader assurance remains incomplete	Conditionally Accepted
CTV-013	GenAI → HealthConnect	Tested emergency scenarios passed after refinement; broader indirect/vague phrasing remains to be assessed	Conditionally Accepted
CTV-014	GenAI → HealthConnect	“I need to be seen ASAP, is that possible?” was treated as scheduling urgency	Validated — Acceptance Pending
CTV-015	GenAI → HealthConnect	Relevant boundary scenarios tested; wider coverage remains subject to further validation	Conditionally Accepted
CTV-016	GenAI → Validation Env.	Two scenarios blocked by Gemini 503/high-demand availability; no pass/fail determination possible	Blocked
CTV-017	GenAI → HealthConnect	Validation remains outstanding	Not Evidenced
CTV-018	All Tracks → Integrated Solution	Integrated workflow has not yet been demonstrated	Pending Validation

Detailed Source Output/Dependency, Alignment Check, Validation Performed and Expected Condition fields for each row are maintained in the full Testing Activity Tracker; this matrix carries the controlled acceptance conclusion.

6. INTERFACE-SPECIFIC VALIDATION REQUIREMENTS

6.1 Analytics ↔ Data Science — Definition and Evidence Alignment

The purpose of this interface is not merely to transfer analytical findings. It is to establish that the analytical basis and modelling basis remain sufficiently consistent for downstream use.

Validation should establish, where applicable:

- Consistent variable definitions;
- Consistent category treatment;
- Consistent interpretation of analytical findings;
- Alignment between relevant analytical summaries and modelling inputs; and
- Documented disposition of material analytical findings.

Current control position: the analytical evidence is available, but receiving-side incorporation/disposition has not been fully evidenced.

Status: Conditional / Acceptance Pending.

6.2 Data Science ↔ ML Engineering — Model and Input Alignment

This is currently the most material cross-track interface. The control objective is to establish that the actual DS-selected candidate, not merely an equivalent stand-in, can be consumed by the ML interface using the agreed feature representation and preprocessing requirements.

Validation must therefore cover:

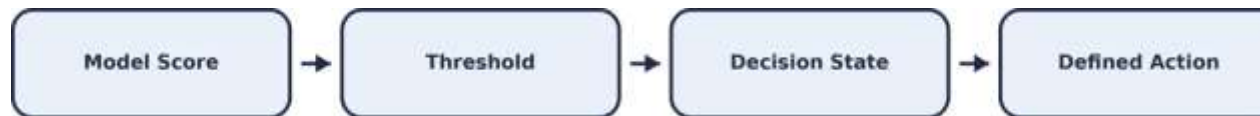
- Candidate identity;
- 33-feature structure;
- Feature ordering;
- Preprocessing assumptions;
- Model interface compatibility;
- Successful execution; and
- Output verification.

Current control position: ML technical interface capability has been demonstrated, but the actual DS candidate and complete input contract remain outstanding.

Status: Red / Acceptance Not Established.

6.3 ML ↔ Operational Decision Pathway

The purpose of this interface is to establish that a technically valid model output can support the next project decision. The control sequence is:



Validation must establish: accepted threshold; decision classification; expected behaviour at the selected threshold; and downstream action associated with the resulting decision state.

Current control position: the model can produce validated scores, but the operational decision pathway remains incomplete.

Status: Red / Acceptance Not Established.

6.4 Decision ↔ Intervention Workflow

This interface establishes whether the model-driven decision has an identifiable downstream response. The control question is:

| What happens after the system identifies the relevant risk state?

Validation should demonstrate that the defined decision state produces a corresponding, testable intervention within the HealthConnect project scope.

Current control position: a complete decision-to-intervention pathway has not yet been demonstrated.

| Status: Red / Acceptance Not Established.

6.5 GenAI ↔ HealthConnect Workflow

The GenAI interface is assessed primarily through behavioural validation, rather than infrastructure architecture. The relevant controls are:

- Knowledge Base grounding;
- Emergency escalation;
- Diagnosis boundaries;
- Treatment/medication boundaries;
- Scheduling-urgency handling;
- Privacy-sensitive interactions;
- Response consistency; and
- Validation availability.

Current control position: selected boundaries have been validated, but broader safety/privacy and availability coverage remains incomplete.

| Status: Amber / Conditional.

7. CROSS-TRACK VALIDATION DIMENSIONS

The following dimensions provide the common validation lens across HealthConnect interfaces. These are control criteria, not assertions that every dimension has already been satisfied.

Dimension	Validation Question
Definition Alignment	Are the relevant variables, metrics, categories and interpretations understood consistently across the interface?
Data / Schema Compatibility	Can the receiving track consume the source output using the intended structure and preprocessing?
Output Integrity	Does the output remain valid and correctly interpretable when transferred downstream?

Dimension	Validation Question
Functional Compatibility	Does the receiving component behave as expected when using the source output?
Operational Alignment	Can a validated technical output support the next defined project decision or action?
Knowledge Grounding	For GenAI, does the response remain within the approved HealthConnect Knowledge Base and information boundary?
Safety Boundary	Does the receiving behaviour remain within the defined emergency, diagnosis, treatment/medication and privacy boundaries?
Availability / Exception Control	Where validation is prevented by service availability, is the condition recorded and appropriately re-executed or dispositioned?
Explainability / Traceability	Can material model behaviour or project decisions be traced to relevant evidence within the assessed scope?
Receiving Acceptance	Has the receiving context confirmed that the output is fit for its intended downstream use?

8. VALIDATION EVIDENCE MATURITY

Cross-track evidence is assessed separately from individual workstream completion.

Level	Definition	Current Position
Level 1 — Output Evidence	Evidence that a workstream produced the required component or result.	Substantially established
Level 2 — Interface Evidence	Evidence that the output was transferred, compatible and usable by the receiving context.	Partially established
Level 3 — Acceptance Evidence	Evidence that the receiving context has formally accepted the output for its intended downstream use.	Incomplete across critical interfaces
Level 4 — Integrated Behaviour Evidence	Evidence that accepted components operate together through the intended HealthConnect workflow.	Not yet established

This maturity model prevents a Level 1 output from being incorrectly represented as Level 3 or Level 4 integration evidence.

9. CROSS-TRACK ACCEPTANCE CRITERIA

An interface may move to Accepted only when the applicable conditions below have been demonstrated:

Output

- Source output is identifiable;
- Intended downstream purpose is defined.

Handoff

- Receiving track has access to the relevant output;
- Transfer/receipt is traceable.

Alignment

- Relevant definitions, structures, assumptions or preprocessing requirements are compatible.

Validation

- Appropriate receiving-side validation has been performed;
- Expected and actual behaviour have been compared.

Resolution

- Material findings have been corrected or formally dispositioned;
- Required re-testing has been completed.

Acceptance

- Receiving context confirms suitability for downstream use;
- Limitations are documented where applicable.

Evidence

- Acceptance is supported by traceable project evidence.

10. CONDITIONAL ACCEPTANCE CONTROL

Conditional acceptance is permitted only where the remaining condition does not invalidate the intended controlled use of the output.

The condition must specify:

- What has been accepted;
- What remains unresolved;
- The operational scope of the acceptance;
- Responsible owner;
- Required closure action; and
- Applicable review/closure point.

A Critical unresolved dependency must not be represented as conditionally accepted merely to improve the apparent readiness position.

11. RETEST AND DISPOSITION CONTROL

Where cross-track validation identifies a material mismatch or failure, the control sequence is:



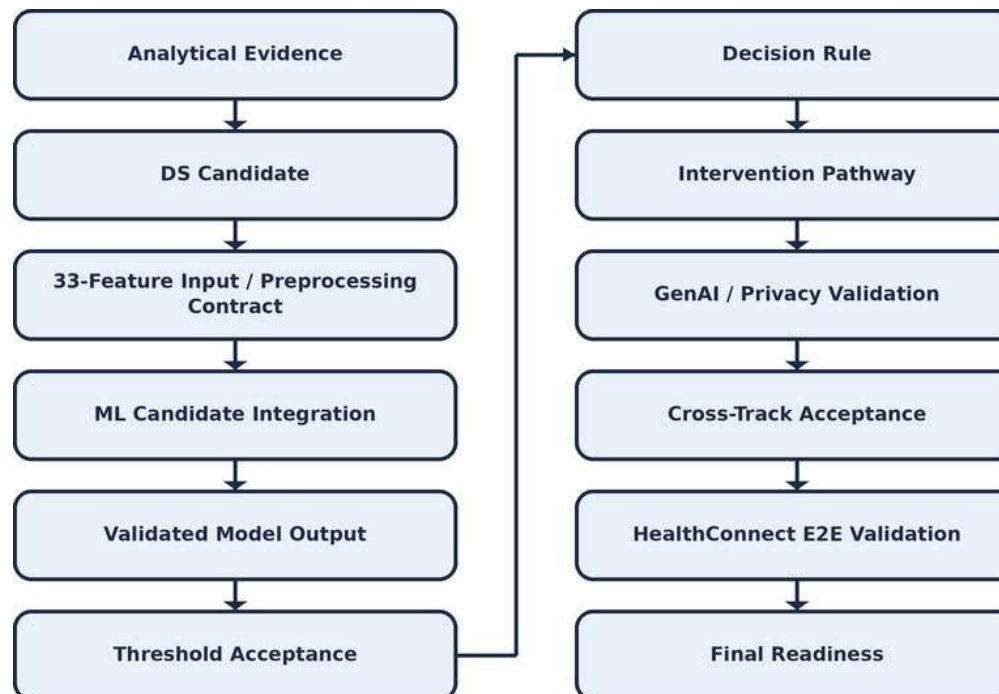
A corrective action alone does not establish closure. Closure requires evidence that:

1. The affected interface has been re-tested;
2. The expected condition has been demonstrated;
3. Downstream impact has been considered;
4. The receiving context remains satisfied; and
5. Any residual limitation has been explicitly dispositioned.

Where an external service prevents execution, the item remains Blocked until it is re-executed or an appropriate formal exception disposition is established.

12. CRITICAL CROSS-TRACK DEPENDENCY PATH

The current HealthConnect acceptance path is:



This sequence is controlled as a dependency chain. Downstream acceptance should not be treated as complete where a material upstream interface remains unverified.

13. CURRENT CROSS-TRACK ACCEPTANCE POSITION

OVERALL STATUS: AMBER – RED

The evidence base demonstrates meaningful cross-track progress, but full cross-track acceptance has not been achieved.

Established Evidence

- Analytics outputs are available for downstream use.
- DS candidate validation has been substantially performed.
- ML technical execution has been extensively validated.
- ML explainability capability has been established.
- Selected GenAI safety and grounding boundaries have been tested.
- Specific GenAI refinements have been re-tested.

Material Acceptance Gaps

- Actual DS-selected candidate → ML integration;
- 33-feature schema/preprocessing reconciliation;
- Model threshold acceptance;
- Score → decision validation;
- Decision → intervention validation;
- Analytics → DS downstream disposition;
- Outstanding GenAI safety/privacy coverage;
- Blocked GenAI scenario execution; and
- Complete HealthConnect E2E validation.

These gaps are not duplicated as full issue narratives here; their corrective control remains in the Integration & Issue Log, Risk & Dependency Register and Decision Log.

14. WEEK 8 CROSS-TRACK CONTROL PRIORITIES

Priority	Required Outcome	Acceptance Evidence
1	Actual DS candidate integrated into ML	Confirmed handoff, input reconciliation and successful execution
2	Model output operationalised	Accepted threshold, decision rule and intervention pathway
3	Analytics interface accepted	KPI reconciliation and downstream disposition
4	GenAI validation gaps closed	Completed scenarios or formal dispositions
5	Cross-track interfaces accepted	Receiving-side confirmation
6	Integrated workflow validated	End-to-end HealthConnect evidence
7	Final acceptance position established	Updated readiness assessment and decision record

15. PM GOVERNANCE AND TRACEABILITY

This matrix is intentionally maintained as a cross-track acceptance artifact, not as a replacement for the project's other control documents.

Artifact	Primary Control Function
Testing Activity Tracker	Individual test execution and retest status
Integration & Issue Log	Material interface issues and corrective actions
Risk & Dependency Register	Project-level exposure and dependency control
Decision Log	Formal decisions, acceptance rationale and consequences
Testing Progress & Status Report	Consolidated testing position
Updated Timeline	Schedule and critical-path implications
Week 8 Readiness Assessment	Overall progression and final readiness determination
Cross-Track Validation & Acceptance Matrix	Evidence that outputs crossed interfaces and became acceptable for downstream use

Traceability principle: this matrix records the acceptance state; the originating control artifact retains the detailed evidence.

This prevents uncontrolled duplication while maintaining an auditable relationship between testing, integration, acceptance and readiness.

16. CONTROLLED CONCLUSION

Week 7 has established substantial component-level validation evidence and meaningful progress across the HealthConnect workstreams.

The remaining control challenge is not simply whether each track has produced an output. It is whether those outputs can be consistently consumed, validated and accepted across the interfaces on which the integrated solution depends.

The most material unresolved interface remains:

Data Science → ML Engineering

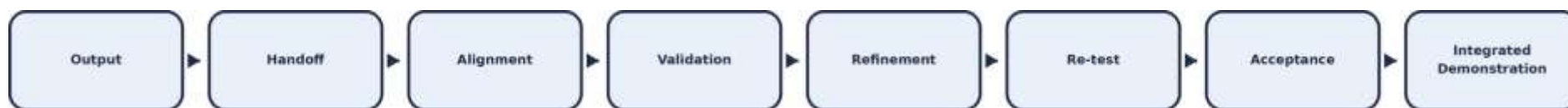
with consequential dependencies extending through: Input Contract → Model Execution → Threshold → Decision → Intervention → E2E Validation.

The GenAI stream presents a separate conditional validation path centred on Knowledge Base grounding, safety boundaries, privacy-sensitive behaviour and execution availability.

Cross-track validation is materially progressed but not complete.

Week 8 should focus on converting the remaining interface evidence into receiving-side acceptance, followed by integrated workflow validation.

The governing progression:



Only after this sequence is sufficiently evidenced should the project move from conditional integration readiness to final integrated acceptance.